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Supplementation of KOH to improve salt tolerance of methanogenesis in the two-stage anaerobic digestion of food waste using pre-acclimated anaerobically digested sludge by air-nanobubble water

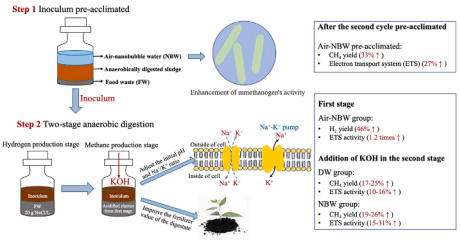
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HIGHLIGHTS

- ETS activity of AD sludge pre-acclimated by Air-NBW was enhanced 3.4 times.
- H_2 yield was increased by 46% in the NBW group.
- Start-up of each stage was accelerated in the Air-NBW pre-acclimated reactor.
- Addition of KOH effectively alleviated the inhibition of salt on methane production.
- Supplementation of K^+ could improve the fertilizer value of the digestate.

GRAPHICAL ABSTRACT



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ABSTRACT

Air-nanobubble water (NBW) was applied to pre-acclimate anaerobically digested sludge that was then used as the inoculum in the two-stage anaerobic digestion (AD) of high saline (20 g NaCl/L) food waste (FW) to optimize NBW application in the AD of high saline FW. K^+ was simultaneously supplemented during the methanogenic stage to resist the inhibition of salt on methanogens. Results showed that after the second pre-acclimation cycle, the inoculum activity was increased 27% in the Air-NBW supplemented reactor in comparison to the deionized water (DW) supplemented one. In the first-stage AD, H_2 yield was enhanced by 46% in the Air-NBW pre-acclimated sludge reactor compared with the DW pre-acclimated sludge reactor. Besides, supplementation of KOH in the methanogenic stage could enhance methane production by 17–25% in the DW reactors at initial pH 7.5, 8.0, and 9.0 when compared to the control reactor (using NaOH adjusted initial pH to 7.5), respectively.

1. Introduction

Salinity is one of the most stressful abiotic factors that limit hydrogen and/or methane production in the anaerobic digestion (AD) process, which has attracted much attention of researchers. On the other hand, food waste (FW) is becoming an increasingly preferred feedstock for AD, which normally contains a high salinity because salt is extensively used as a food flavoring agent, especially in Japan. Asakura et al. (2014) pointed out that a much higher level of Na^+ than the world health organization (WHO) recommendation was consumed by most Japanese due to the Japanese diet. As known, high salinity (>10 g NaCl/L) usually increases the osmotic pressure outside the cells and causes the outflow of intracellular water or even the cell rupture, which strongly inhibits the metabolism of non-halophilic bacteria (Li et al., 2018). Therefore,

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excessive salt present in organic wastes, i.e., FW, could inhibit the activities of microorganisms in all stages (hydrolysis/acidification and methanogenesis) of AD to varying degrees, resulting in a lower AD performance and even process failure. Among them, high salinity may more remarkably inhibit methane production than hydrogen and volatile fatty acids (VFAs) production because the methanogens are more sensitive to environmental factors like high salt concentration than the hydrolytic bacteria (Wang et al., 2018a). Additionally, it has been found that anaerobic microbes need some time to accommodate the hypersaline environments when dealing with the highly saline FW, which is probably the main reason for a longer startup time (about 20 days) at 20 g NaCl/L in a previous study (Hou et al., 2021b). Hierholtzer and Akunna (2014) also found a similar phenomenon that methane production was delayed by about 15 days in the AD of saline-rich macroalgae system. These observations suggest that high salinity could strongly inhibit microbial activity during the early stage of methanogenesis. Therefore, some strategies for addressing this problem though improving microbial activity are recommended.

In general, the tolerance of microorganisms to high salinity could be improved by the synthesis of themselves or adding external osmoprotectants, allowing microorganisms to accumulate osmolytes (compatible solutes) and balance their cytoplasms (Salma et al., 2020). Amino acids and quaternary ammonium compounds, including proline, choline, glycine betaine, carnitine, and trehalose, have been adopted as common osmoprotectants in the AD of high saline organic wastes to recover methanogenic activity and increase methane production (Oh et al., 2008; Liu et al., 2019b; Zhang et al., 2016). However, Liu et al. (2019b) reported that excessive addition of osmoprotectants had an adverse effect on methane production. Apart from this, the osmoprotectants are organic substances that may be consumed by microorganisms as carbon or nitrogen sources, resulting in instability in continuous AD systems. Therefore, seeking other methods to improve the methanogens activity in a high salt environment is necessary. Most recently, some researchers had reported that the nanobubble water (NBW) (the water containing bubbles with the diameter <1000 nm) has the unique characteristics including negative charge on their surfaces, high mass transfer efficiency, and large specific surface area of NBs, which may facilitate the chemical reaction, physical adsorption, and mass transport at the gas-liquid interface (Gurung et al., 2016). Therefore, adding NBW in AD system as a novel strategy has been put forward and demonstrated to enhance the bioactivities of enzymes involving in the AD systems, including hydrolases, cellulase, and coenzyme F_{420} (Hou et al., 2021a; Hou et al., 2021b; Wang et al., 2020a). According to previous studies, however, the volume of NBW-based AD reactors is relatively large, which could increase the construction costs. On the contrary, if the anaerobically digested sludge is firstly acclimated by NBW and then used as the inoculum to the AD system, not only the activity of microorganisms can be enhanced, but also the volume of reactors may be largely reduced when compared to that with NBW being added directly.

On the other hand, salinity can induce K^+ efflux from cells, causing osmotic pressure imbalance and ultimately cell lysis and death because K^+ is an essential element for regulating intracellular osmotic pressure (Larbi et al., 2020; Plemenitas et al., 2014). Meanwhile, K^+ as a favorable alternative osmolyte in cells can be transported through the unique potassium pump system on the cell membrane (Zhang et al., 2016). Therefore, supplementation of K^+ as an accelerator may provide a new train of thought for improving the biological treatment process of high saline wastes. Gul et al. (2019) reported that the moderately supplemented K^+ reduced the concentration of Na^+ in tissues and cells and alleviated the deleterious effects of Na^+ toxicity on the wheat growth and yield. Gan et al. (2021) also found that the proportion of genus *Lactobacillus* was increased from 22% (fresh meat) to 73%, 82%, 77% and 86% at the end of processing when NaCl was partially replaced with 0%, 30%, 50%, and 70% of KCl. It is noteworthy that addition of 0.174% KCl to the AD of saline organic wastewater increased methane production by about 49% when the NaCl content was 2.0% (Li et al., 2018). The residual digestate slurry at the end of hydrolysis/acidification stage is always weakly acidic from the two-stage AD of FW. When it is used as the substrate for the methanogenic stage, proper adjustment of pH value using bases such as NaOH is necessary in order to meet the suitable growth environment for methanogens. Therefore, substitution of NaOH with KOH as a pH regulator may not only reduce the additional increase in Na^+ but also mitigate the toxicity of high salinity on methanogens during the AD of high saline organic wastes due to the supplemented K^+.

This study was conducted in two steps to eliminate the adverse effects of high salt concentration (20 g NaCl/L) on two-stage AD of FW. Firstly, the anaerobically digested sludge was acclimated by using Air-NBW or deionized water (DW) (as control) and then the acclimated sludge was used as inoculum during the two-stage AD of FW system. Secondly, KOH was added to adjust the pH at the beginning of methanogenic stage to 7.5, 8.0, and 9.0, respectively, and then the effects of different K^+ dosages on methanogens under the high salt concentration by using the DW or Air-NBW pre-acclimated anaerobically digested sludge were examined. This study is expected to develop a feasible strategy for the practical application of Air-NBW and offer novel insights into the improvement of salt tolerance of anaerobic microorganisms in the two-stage AD of high saline organic wastes.

2. Materials and methods

2.1. Anaerobically digested sludge, feedstock, and Air-NBW

Anaerobically digested sludge was gained from the anaerobic bioreactor of Chikusai wastewater treatment center in Ibaraki, Japan. The initial total solid (TS) and volatile solid (VS) of the raw sludge were 0.88 ± 0.04% and 0.58 ± 0.05%, indicating its high moisture content (Table 1). Therefore, the sludge was concentrated through a 1 mm pore size sieve and centrifuged at 4000 rpm for 15 min. The TS and VS of the concentrated sludge were 2.45 ± 0.02% and 1.92 ± 0.02%. In this study, synthetic FW was used, which was the same as reported in a previous study (Hou et al., 2021b). The main physicochemical characteristics of FW and raw anaerobically digested sludge are also described in Table 1. For the manufacturing process of Air-NBW, a NBW generator (HACK FB11, Japan) with the pressure of 0.22–0.25 MPa and the flow rates of gas and water of 0.05 L/min and 3 L/min was used, which was based on the previous study (Hou et al., 2021b). After 20 min generation time, the number of bubble particles was around 3×10^8 particles/mL with the particle size ranging from 59 to 127 nm.

2.2. Pre-acclimation of anaerobically digested sludge by DW or Air-NBW

Inoculum plays a pivotal role in the anaerobic biological process because the stability and efficiency of AD have a strong correlation with the number, species, and metabolic activities of microorganisms (Franchi et al., 2018). In this study, the anaerobically digested sludge with the initial electron transport system (ETS) activity of 19.87 ± 0.63 mg/(g·h) was pre-acclimated by Air-NBW, which was expected to improve the microbial activity and help microorganisms resist the toxicity of high salt.

For the pre-acclimation process, 200 mL concentrated sludge and 10

Table 1
The main physicochemical characteristics of food waste and raw anaerobically digested sludge.

Parameter	Food waste	Raw sludge
Total solids (TS, %)[a]	18.77 ± 0.40	0.88 ± 0.04
Volatile solid (VS, %)[a]	18.01 ± 0.03	0.58 ± 0.05
Total volatile fatty acids (VFAs, mg/L)	1211.16 ± 5.77	43.21 ± 0.19
Lactic acid (g/L)	4.30 ± 0.17	/
Soluble proteins (mg/L)	13810.88 ± 112.77	295.51 ± 8.90
Soluble carbohydrates (mg/L)	23100.80 ± 86.55	22.66 ± 0.45

[a] Wet weight basis.

g FW (wet weight) at a substrate to inoculum ratio (S/I) (VS basis) of 0.5 were added into each 500 mL digester, and then DW or Air-NBW was used to adjust its working volume to 350 mL. Each reactor was purged by pure N_2 for 2 min, 3 times and the initial pH was 6.79 ± 0.42 in the first cycle and 7.51 ± 0.12 in the second cycle with no additional adjustment. All the digesters were placed at a constant temperature of 38 ± 1°C. At the end of test, the reaction slurry was concentrated again through the 1 mm pore size sieve and centrifuged at 4000 rpm for 15 min, which was then utilized as the inoculum in the second cycle to repeat the above acclimation procedure. During the two cycles of acclimation, methane production was measured every two days and ETS activity was detected at the end of each cycle (around one month).

2.3. Two-stage AD of high saline FW by using Air-NBW pre-acclimated anaerobically digested sludge as inoculum

2.3.1. Performance of the first-stage AD

Hydrogen production potential tests were carried out in 100 mL reactors, and the working volume of each bottle was 55 mL. 15 g (wet weight) of FW together with the 20 g/L of salt (NaCl) concentration and 40 mL of above pre-acclimated anaerobically digested sludge as inoculum were added to each reactor. A higher initial total VS about 6% was achieved in each reactor, and the S/I ratio was approximately 6.0 (VS basis). No additional chemical reagents were added to adjust the pH during the first stage. Although FW had a low pH of 4.30, the initial pH in each reactor was 6.50 ± 0.20 after all the above procedures being completed, which was within the appropriate pH range for hydrogen production. The reactors were labelled as DW and Air-NBW according to the water added during the pre-acclimation stage.

2.3.2. Effect of KOH addition on the performance of the second-stage AD

For methane production potential tests, the reactors with the same specifications as in the first stage were used with the same working volume of 55 mL each. The residual acidified slurry from the first stage and the pre-acclimated anaerobically digested sludge were respectively added as substrate and inoculum in each reactor (at the S/I ratio of 2 (VS basis)). The initial total VS in each reactor was 1.9 ± 0.02%. Afterwards, in order to evaluate the effects of KOH as an accelerator on methane production from the AD of high saline FW, the initial pH of each reactor was adjusted to 7.5, 8.0, and 9.0 using 6 M KOH, respectively. The initial pH of the control group was adjusted to 7.5 by using 6 M NaOH. And these reactors were designated as $DW_{control}$-7.5, DW-7.5, DW-8.0, DW-9.0, $NBW_{control}$-7.5, NBW-7.5, NBW-8.0, and NBW-9.0, respectively.

Each reactor was hermetically sealed and purged with pure N_2 for 2 min, 3 times to maintain the anaerobic condition in all experiments prior to start and then placed in a constant temperature incubator (38 ± 1°C). Biogas production was measured regularly at room pressure (25°C, 1 atm) and the contents of hydrogen and methane were checked by a gas chromatograph (GC-8A, Shimadzu, Japan).

2.4. Analytical methods and calculations

The contents of TS, VS, and alkalinity including total alkalinity (TA), partial alkalinity (PA, endpoint pH to 5.75), and intermediate alkalinity (IA, pH from 5.75 to 4.30) were based on the standard methods (APHA, 2012). A semi-solid pH meter (Testo 206, Germany) was used to monitor the pH value. In addition, the liquid samples were centrifuged at 9900 rpm for 20 min and filtrated through 0.22 μm membrane filters to determine the concentrations of soluble protein, soluble carbohydrate, VFA, and lactic acid. Among them, the concentration of soluble protein, carbohydrate, and VFAs were determined according to a previous study (Hou et al., 2020). Lactic acid was quantified by using a spectrophotometer (Shimadzu, Japan) according to Borshchevskaya et al. (2016). For the determination of concentrations of NH_4^+ (total ammonia nitrogen, TAN) and K^+, the samples were filtered through 0.22 μm filters and then measured by Ion Chromatography (IC, Shimadzu, Japan). Free ammonia nitrogen (FAN) was calculated according to the following Eq. (1) (Zhao et al., 2020a).

$$FAN = \frac{TAN \times 10^{pH}}{e^{6344/(273+T)} + 10^{pH}} \quad (1)$$

where T is the temperature in Celsius (38°C in this study).

The microbial activities during the two-stage AD were reflected by the electron transport rate, i.e., ETS activity. The 2-(p-iodophenyl)-3-(p-nitrophenyl)-5-phenyltetrazolium chloride-electron transport system (INT-ETS) reduction method was used. The specific procedures were based on the report by Wang et al. (2020a). 0.1 mL of 0.2% INT solution was added to 0.5 mL of samples, and the centrifuge tubes were incubated in a water bath oscillator at 100 rpm and 37 ± 1°C for 30 min under dark conditions. The reaction was then terminated with 1 mL of 37% methanol, and the reactants were centrifuged at 10000 rpm, 4°C for 5 min before the supernatant was discarded. Subsequently, 5 mL of 98% methanol was added to the sediment to shake in the same water bath oscillator again for 10 min under a dark environment. Then, the supernatants were obtained after centrifugation (10000 rpm, 4°C for 5 min) and determined by a spectrophotometer (Shimadzu, Japan) at 485 nm. Finally, the sediments were dried to constant weight at 105 ± 1°C. The kinetic analysis by using the modified Gompertz model was carried as mentioned previously (Hou et al., 2021a).

2.5. Data calculation and analysis

All the reactors operation were performed in duplicate with triplicate measurements for each sample. One-way analysis of variance (ANOVA) was conducted to probe the statistical difference via Microsoft Office Excel 2010, and statistical significance was assumed at $p < 0.05$.

3. Results and discussion

3.1. CH_4 yield and ETS activity during the pre-acclimation process

Microorganisms with robust physiology are essential for the stable performance of AD. When fed with a new substrate, the established consortium of anaerobically digested sludge generally needs time to adapt to the new environment (Vásquez and Nakasaki, 2016). When FW is directly used as the substrate to the AD system, the imbalance occurred would lead to a slow start-up with a low efficiency (Zeng et al., 2019). Therefore, some pre-acclimation of anaerobically digested sludge is necessary. In this study, the anaerobically digested sludge was pre-acclimated at a low S/I ratio by adding Air-NBW, so as to enhance the microbial activities of the inoculum and the performance of the two-stage AD of high saline FW.

During the two cycles of pre-acclimation, the cumulative CH_4 yields were examined, and the results are displayed in Fig. 1a. As seen, the cumulative CH_4 yields in the Air-NBW reactor (169.88 ± 1.83 mL/g-VS_{added} in the first cycle and 217.46 ± 0.50 mL/g-VS_{added} in the second cycle) were higher around 14% and 33% than that in the DW reactor during both cycles, respectively. ETS activity also showed the similar results, which were increased by 0.9 and 1.2 times in the first cycle and 2.5 and 3.4 times in the second cycle in the DW and Air-NBW pre-acclimated sludge reactor when compared to the raw anaerobically digested sludge (19.87 ± 0.63 mg/(g·h)), respectively (shown in Fig. 1b). This observation indicates that pre-acclimation of anaerobically digested sludge is an effective enhancement approach for microbial activity. After the second cycle, the highest ETS activity was detected in the Air-NBW reactor (87.03 mg/(g·h)), increasing by 27% when compared with that in the DW reactor. Xiao and Xu (2020) also found that use of NBs technology would achieve 6 times higher dehydrogenase activity than that of coarse bubbles, resulting in enhanced microbial metabolism and microorganisms' proliferation. This finding agrees with Gurung et al. (2016) who stated that NBs technology possesses greater

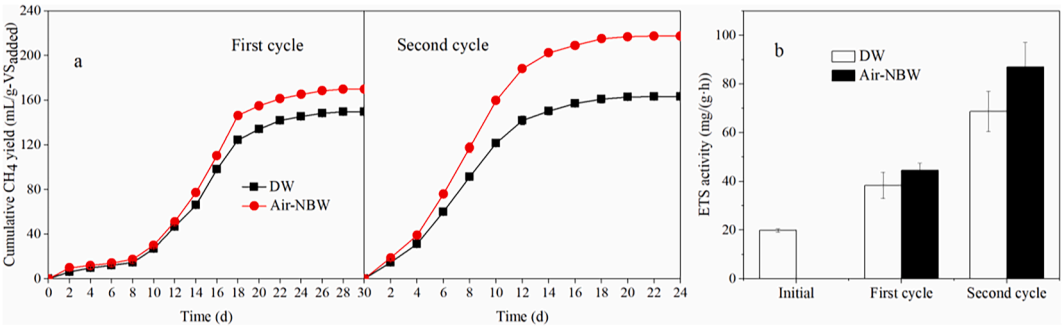


Fig. 1. (a) Cumulative CH_4 yield during the two-cycle pre-acclimation, and (b) changes in electron transport system (ETS) activity at the end of each cycle during the pre-acclimation by DW or Air-NBW.

application potentials in the field of environmental engineering because it can enhance biological activity and mass transfer efficiency to a great extent. Results from this study also suggested that anaerobically digested sludge pre-acclimated by Air-NBW exhibited a higher biological activity and methane production potential.

3.2. Reactor performance at hydrolysis/acidification stage by using Air-NBW pre-acclimated anaerobically digested sludge as inoculum

3.2.1. H_2 yield and ETS activity

The hourly H_2 production, cumulative H_2 yield, and ETS activity during the first stage are displayed in Fig. 2. The H_2 production was completed within 48 h, indicating a quick start-up of the reactors. In particular, the hourly H_2 production from the Air-NBW group was higher than that from the DW group during the whole stage, and the highest value was obtained at 12 h, about 29.45 ± 1.21 mL/g-VS_{added}. A previous study (Hou et al., 2021b) showed a lag time (λ) of 17.87 ± 1.30 h at 20 g NaCl/L. Cao and Zhao (2009) also obtained the λ of 14.6 and 29.2 h when the salt concentrations were 19.19 and 24.49 g NaCl/L, respectively. However, in this study no inhibition was observed at the high salinity of 20 g NaCl/L when the pre-acclimated sludge was used as inoculum. The λ was only 3.72 ± 0.06 h in the DW pre-acclimated sludge reactor and 2.90 ± 0.02 h in the Air-NBW pre-acclimated sludge reactor, respectively (Table 2). As shown in Fig. 2b, the cumulative H_2 yield from the Air-NBW group was enhanced by 46% in comparison to the DW group. These data confirm that the performance of H_2 production could be enhanced at the high salinity condition in the AD reactor inoculated with the Air-NBW pre-acclimated sludge.

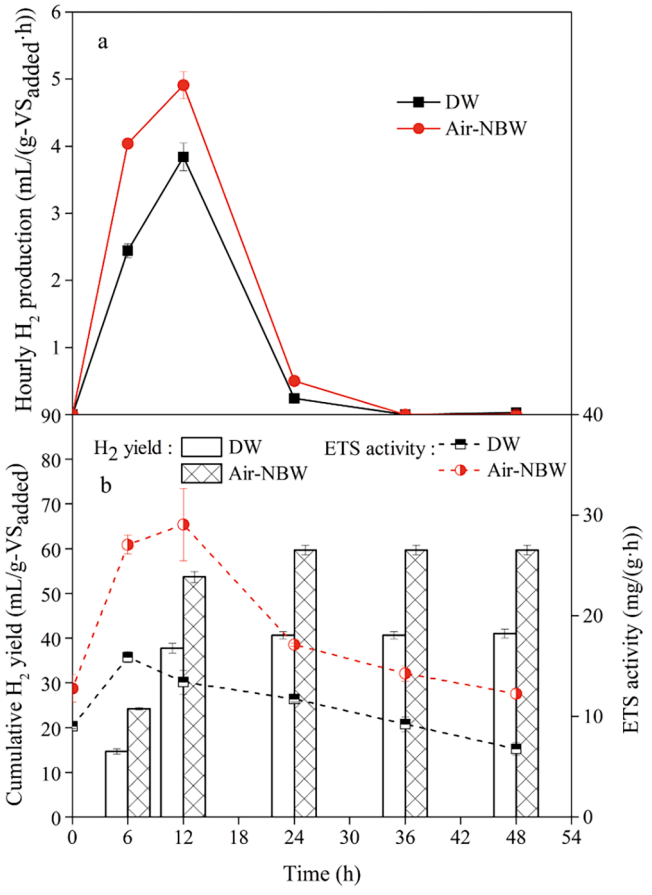


Fig. 2. Hourly H_2 production (a), cumulative H_2 yield and ETS activity (b) during the first stage AD of high saline (20 g NaCl/L) FW inoculated with anaerobically digested sludge pre-acclimated by DW or Air-NBW.

The changes of ETS activity at the first stage (Fig. 2b) were consistent with the results of hydrogen production, which can indirectly reflect the metabolic activity relating to electron transfer in microbial cells. High hydrogen production was accompanied by an increase in ETS activity, and the ETS activity in the Air-NBW group was always higher than that in the DW group. In particular, the ETS activities in the Air-NBW

Table 2
Kinetic analysis of H_2 and CH_4 yield from the first stage and second stage AD tests with the addition of KOH.

Stage/ Reactors	Modified Gompertz model			
First-stage				
	P_{H2} (mL/g-VS_{added})	$R_{H2,\ max}$ (mL/(g-VS_{added}·h))	λ (h)	R^2
DW	40.76 ± 0.09	17.52 ± 0.40	3.72 ± 0.06	0.9999
Air-NBW	59.71 ± 0.03	21.26 ± 0.09	2.90 ± 0.02	1.0000
Second-stage				
	P_{CH4} (mL/g-VS_{added})	**$R_{CH4,\ max}$ (mL/(g-VS_{added}·d))**	**λ (d)**	**R^2**
$DW_{control}$-7.5	226.88 ± 3.08	68.43 ± 3.07	12.25 ± 0.20	0.9980
DW-7.5	276.27 ± 2.67	72.42 ± 3.77	11.03 ± 0.09	0.9986
DW-8.0	281.41 ± 2.49	74.56 ± 2.75	9.38 ± 0.09	0.9985
DW-9.0	261.86 ± 2.61	70.99 ± 2.49	9.09 ± 0.18	0.9986
$NBW_{control}$-7.5	237.32 ± 3.17	70.24 ± 2.10	9.69 ± 0.14	0.9962
NBW-7.5	282.17 ± 3.06	73.26 ± 5.65	9.27 ± 0.11	0.9978
NBW-8.0	299.52 ± 2.50	75.15 ± 2.75	8.41 ± 0.22	0.9977
NBW-9.0	295.58 ± 3.19	81.08 ± 3.13	9.14 ± 0.19	0.9983

reactors were 0.7 and 1.2 times higher than the DW reactor at 6 and 12 h, respectively, suggesting the more excellent microbial metabolic activity of the Air-NBW pre-acclimated sludge. This may be attributable to the increased microbial biomass after being acclimated by Air-NBW. Zhao et al. (2020b) reported the increase in the diversity and abundance of archaea and bacterial communities after acclimation. More importantly, Xiao and Xu (2020) found that the NBs could significantly accelerate the formation of biofilm and increase the biofilm thickness.

3.2.2. Variation in VFAs and lactic acid concentration during the hydrogen fermentation

Besides pH, the concentrations of VFAs and lactic acid were detected during the first stage (Fig. 3). Previous research proved the existent of different hydrogen-producing fermentation pathways (i.e., butyric acid (HBu)-type fermentation and mixed-acid fermentation) in anaerobic fermentation systems, which may lead to the accumulation of various end products (Lee et al., 2012). VFAs as the main by-product from H_2 production were accumulated rapidly in the first 12 h and after that a slow increase was observed (Fig. 3a). At the end of 48 h, the total VFA concentrations were 4189.30 ± 37.94 and 4948.41 ± 177.26 mg/L in the DW reactor and Air-NBW reactor, respectively. The concentration of HBu was about 33% higher in the Air-NBW reactor than that in the DW reactor, suggesting that the HBu-type fermentation was more dominant in the Air-NBW reactor. In addition, a high carbohydrates content (41–62%) in FW often leads to the accumulation of lactic acid, which is the main product of carbohydrate fermentation (Anna et al., 2013). However, the lactic acid concentration was not significantly different among the reactors during the first 6 h, and from 12 h on its concentration in the DW reactor was always higher than that in the Air-NBW reactor (Fig. 3b). At the end of the first stage, the lactic acid concentrations were 9.50 ± 0.20 and 7.64 ± 0.16 g/L in the DW reactor and Air-NBW reactor, respectively. This indicates that the microbial community might shift toward lactic acid metabolism in the DW reactor, probably attributing to the decrease in hydrogen production. Ghimire et al. (2018) investigated the accumulation of lactic acid during hydrogen production and found that the decrease in hydrogen production was related to the metabolic transfer caused by lactic acid accumulation. Anna et al. (2013) summarized the reasons for the inhibition of hydrogen production by lactic acid, including the competition of lactic acid bacteria for substrates and the excretion of bacteriocins that can inhibit the growth of other microorganisms. In addition, the accumulation of lactic acid also contributed to a lower pH in the DW group (Fig. 3b). At the end of the first stage, the pH values were 3.83 ± 0.01 and 3.94 ± 0.03 in the DW group and the Air-NBW group, respectively.

3.3. Effects of KOH addition on CH_4 production from second-stage AD inoculated with Air-NBW pre-acclimated anaerobically digested sludge

3.3.1. CH_4 yield and ETS activity

The CH_4 production rate and cumulative CH_4 yield during the second stage are displayed in Fig. 4. As shown, there is no significant difference in CH_4 production among all the reactors during the first 6 days ($p = 0.38 > 0.05$). Notably, the reactors with the addition of KOH were started up faster than the control group, and the λ in the Air-NBW pre-acclimated sludge reactors were shorter than those in the DW pre-acclimated sludge reactors (Table 2). The CH_4 production potential (P_{CH4}) and CH_4 production rate ($R_{CH4,\ max}$) reflected the consistent results, with the highest P_{CH4} and $R_{CH4,\ max}$ being obtained in the NBW-8.0 and NBW-9.0, respectively. As expected, the final cumulative CH_4 yields (Fig. 4b) were about 22%, 25%, and 17% higher in the DW-7.5 (268.54 ± 1.83 mL/g-VS_{added}), DW-8.0 (275.39 ± 2.85 mL/g-VS_{added}), and DW-9.0 (258.09 ± 4.12 mL/g-VS_{added}) than that in the $DW_{contorl}$-7.5 (220.83 ± 1.44 mL/g-VS_{added}), respectively. Correspondingly, the final cumulative CH_4 yields were 19%, 26%, and 25% respectively higher in the NBW-7.5 (281.40 ± 1.12 mL/g-VS_{added}), NBW-8.0 (297.21 ± 3.86 mL/g-VS_{added}), and NBW-9.0 (295.87 ± 7.29 mL/g-VS_{added}) than that in the $NBW_{contorl}$-7.5 (236.79 ± 10.30 mL/g-VS_{added}). These results indicate that the addition of K^+ can effectively alleviate the inhibition of high salinity on methane production. This observation may be associated with the fact that the supplementation of K^+ could regulate the osmotic pressure in microbial cells and prevent the accumulation of Na^+ in the cells so as to maintain a high cell survival rate (Tang et al., 2018). Additionally, a decline trend of methane yield was noticed with the increase of K^+ dosage in the DW pre-acclimated sludge reactors, in agreement with Li et al. (2018) who found that excessive K^+ had a negative effect on methane production. This phenomenon may be brought about by the toxicity of excessive salt concentration to microbial activity (Ashfaq et al., 2020). However, the decline of methane yield was not obvious when the K^+ dosage was increased by adjusting the pH to 9.0 in the NBW pre-acclimated reactors, indicating that the microorganisms would have a stronger tolerance to environmental toxicants after being acclimated by Air-NBW. This may be due to the increased microbial activity after the anaerobically digested sludge pre-acclimated by Air-NBW, which was consistent with the results of ETS activity (shown in Fig. 4c).

As shown in Fig. 4c, the ETS activity in all the reactors was increased gradually during the first 24 days, in agreement with the methane production. It was clearly observed that the ETS activities in the K^+ addition reactors were always higher than that in the control group, in which the ETS activities of the NBW pre-acclimated sludge reactors were also always higher than the corresponding DW group reactors. In the DW

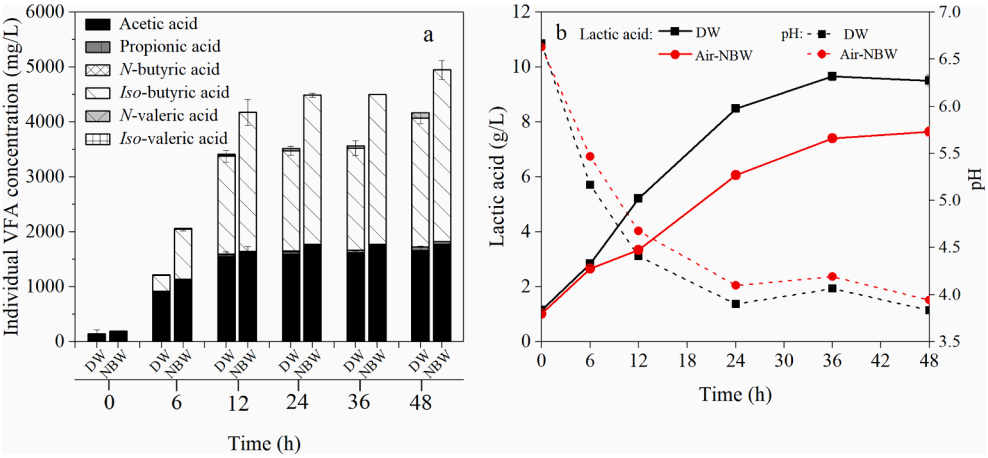


Fig. 3. Changes of the individual VFA concentration (a) and lactic acid and pH (b) during the first stage.

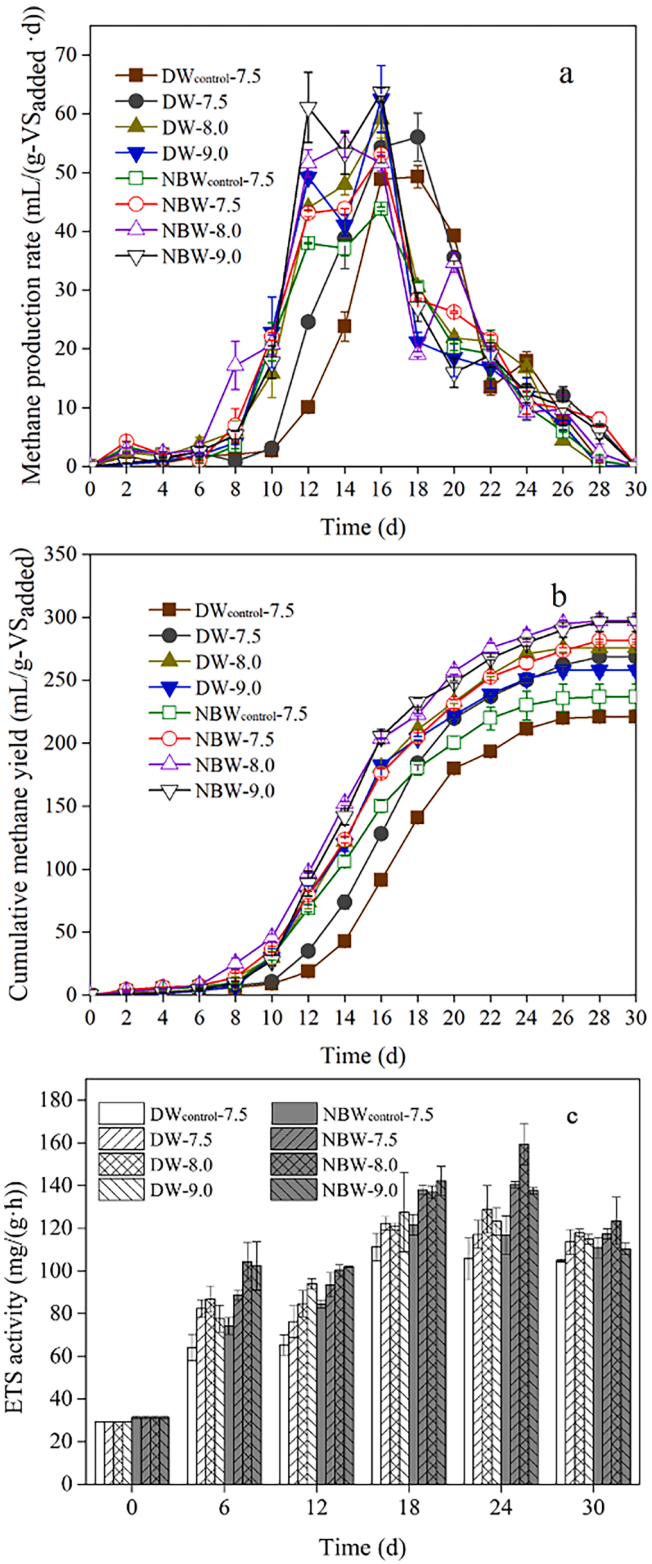


Fig. 4. CH_4 production rate (a), cumulative CH_4 yield (b), and ETS activity (c) during the second stage at different initial pH values adjusted by KOH.

group, the maximum ETS activities were enhanced by 10%, 16%, and 15% in the DW-7.5, DW-8.0, and DW-9.0 when compared with the $DW_{control}$-7.5; in the NBW group, the maximum ETS activities were increased by 15%, 31%, and 17% in the NBW-7.5, NBW-8.0, and NBW-9.0 in comparison to the $NBW_{control}$-7.5. Compared with the DW group, the NBW reactors exhibited 9–23% increase in ETS activity. Results from this study demonstrate that the supplementation of K^+ could improve the activity of microorganisms in the second-stage AD of high saline FW inoculated with the Air-NBW pre-acclimated sludge, to some extent tackling low metabolic activity issue under high salt concentrations. Gagliano et al. (2017) concluded that the addition of K^+ (0.7 g/L) could alleviate the negative effect of high salinity (20 g Na^+/L) on the biofilm formation and the performance of Up-flow Anaerobic Sludge Blanket (UASB) reactor. This observation also agrees with Wang et al. (2020b) who found that the methane yields from the corn straw were increased by 10% and 22% after the sludge being pre-augmented by N_2-NBW and O_2-NBW.

3.3.2. *Reduction of dissolved organic matters during the second stage*

Dissolved organic matters as the important intermediate products, including the soluble proteins, soluble carbohydrates, lactic acid, and individual VFAs, are monitored and showed in Fig. 5. During the first 6 days, the concentration of soluble proteins was decreased slightly and then increased in the DW-9.0 and NBW-9.0. However, the concentrations of soluble carbohydrates and lactic acid were decreased rapidly, especially in the NBW group, which was accompanied with a large accumulation of VFAs. As known, lactic acid is one of the main by-products in the anaerobic fermentation of FW, and its initial hydrolytic product is propionic acid (HPr), which is detrimental to methanogens (Cheng et al., 2020; Li et al., 2014). In this study, the higher accumulation of lactic acid in the DW group (when compared to the NBW group) during the first stage led to the higher initial lactic acid concentration in the second stage (Fig. 5c). This would be accompanied with a higher HPr in the DW group, probably attributable to the lower methane yield in the DW group (Fig. 5d). On day 24, HPr was the dominant VFA in the DW group, about 1229.29 ± 15.39, 865.49 ± 0.83, 584.93 ± 28.63, and 892.21 ± 11.57 mg/L in the $DW_{control}$-7.5, DW-7.5, DW-8.0, and DW-9.0; while acetic acid (HAc) dominated the NBW group, about 873.93 ± 22.88, 787.09 ± 75.45, 607.87 ± 34.13, and 202.75 ± 5.90 mg/L in the $NBW_{control}$-7.5, NBW-7.5, NBW-8.0, and NBW-9.0, respectively.

Moreover, in the K^+ added reactors, the degradation of organic matters was higher than that in the control group, suggesting the higher methane yield in the K^+ added reactors which may be benefited from the rapid consumption of organic matters, especially VFAs. Li et al. (2018) found that at 2.0% NaCl condition the chemical oxygen demand (COD) removal efficiency, dehydrogenase activity, and the viability of microorganisms were enhanced by 115.4%, 77.2%, and 20.3% with the addition of KCl to AD of organic wastewater. Muñoz Sierra et al. (2018) also pointed out that a higher K^+/Na^+ ratio could achieve a higher COD removal efficiency and methane production rate when treating high saline phenolic wastewater, highlighting the importance of K^+ to maintain the methanogenic activity under high salinity environment.

3.3.3. *Variations of pH and alkalinity during the second stage*

During the CH_4 production stage, the initial pH was adjusted to 7.5, 8.0, and 9.0 by using NaOH or KOH, which would inevitably cause the change of alkalinity during the process. Importantly, alkalinity is one of the key factors to maintain the stability of AD operation, which is a better indicator of imbalanced AD process than pH (Wang et al., 2018b). In addition, IA/PA and VFA/TA ratios are also commonly used as the preferable indicators that can monitor the methanogenic process (Sun et al., 2019). Therefore, the changes of pH, TA, IA/PA, and VFA/TA are recorded during the second stage (Fig. 6). The changes in pH were consistent with the results of VFAs accumulation and consumption, i.e., the pH dropped to the lowest when VFAs were largely accumulated, and then the pH was increased with the consumption of VFAs (Fig. 6a). Certainly, the system pH value returned to the optimum pH range (7.5–8.0) for the methanogens more quickly when a higher initial pH was adopted, and this phenomenon was more obvious in the DW group reactors. For instance, the pH values varied from 6.38, 6.48, 6.63, 6.85, 6.51, 6.51, 6.73, and 6.82 to 6.9, 7.04, 7.12, 7.42, 7.45, 7.41, 7.51, and 7.52 in the $DW_{control}$-7.5, DW-7.5, DW-8.0, DW-9.0, $NBW_{control}$-7.5,

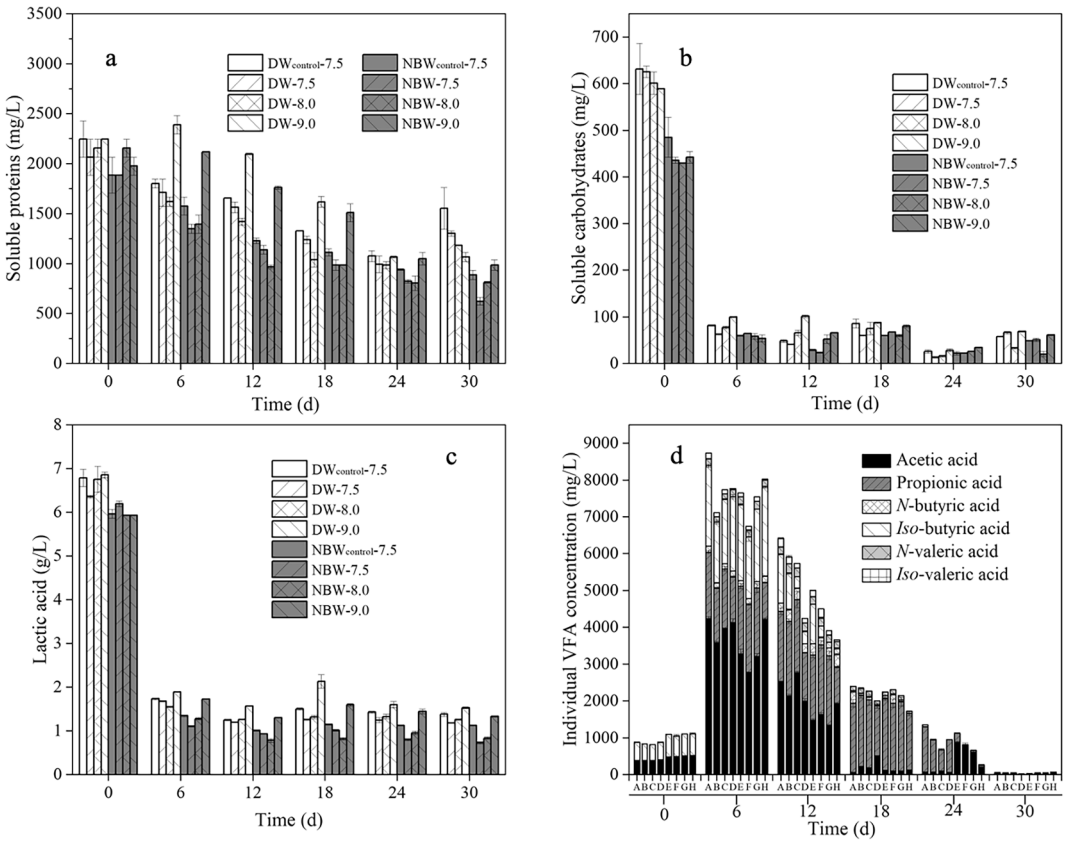


Fig. 5. Variations of soluble proteins (a), soluble carbohydrates (b), lactic acid (c), and individual VFA concentration (d) (The alphabets under X-axis areas: A-DW$_{control}$-7.5; B-DW-7.5; C-DW-8.0 D-DW-9.0; E-NBW$_{control}$-7.5; F-NBW-7.5; G-NBW-8.0; H-NBW-9.0) during the second-stage AD of high saline (20 g NaCl/L) FW.

NBW-7.5, NBW-8.0, and NBW-9.0 on day 6 to day 12, respectively. This observation indicates that addition of KOH plays a better pH regulating role in the DW group, which may be one of the reasons for the increase in methane production. The profiles of TA evolution are displayed in Fig. 6b. A more remarkable TA increase was noticed in the prophase of methane production in all the reactors, likely associated with the degradation of protein-rich substrates responsible for alkalinity production (Adou et al., 2020). This observation agrees with the changes in the concentration of soluble protein, while the increase of TA was only 825.83 and 938.44 mg $CaCO_3$/L in the DW-9.0 and NBW-9.0 on day 6, which was lower than other reactors most probably due to the lower degradation of soluble proteins. However, the TA concentrations in both reactors were increased significantly after day 6, possibly due to the addition of a large number of alkaline matters (6 M KOH).

According to previous studies, PA and IA can be used to distinguish the bicarbonate alkalinity and VFA alkalinity; and the IA/PA ratio is better for monitoring the buffering capacity of the AD process (Jantsch and Mattiasson, 2003; Sun et al., 2019). In the DW$_{control}$-7.5, the IA/PA ratio reached to 4.85 on day 6 (Fig. 6c), indicating its extreme instability because a sudden increase in the IA/PA ratio usually reflects a decrease in pH buffering capacity (Banks et al., 2012). Interestingly, the IA/PA ratio in the NBW group with the addition of KOH were lower than that in the corresponding DW group. This indicates that the buffering capacity of the NBW group was higher than the DW group. As the AD progressed, the IA/PA ratios continued to decline in all reactors, which were decreased to 0.25, 0.24, 0.24, 0.20, 0.17, 0.13, 0.08, and 0.16. in the DW$_{control}$-7.5, DW-7.5, DW-8.0, DW-9.0, NBW$_{control}$-7.5, NBW-7.5, NBW-8.0, and NBW-9.0 at the end of the second stage, respectively. This observation implies that the reactors gradually recovered and became stable. Similarly, the VFA/TA ratio showed an increase trend at first and then decreased in all reactors (Fig. 6d). Liu et al. (2019a) summarized the range of VFA/TA ratios for the stable operation of AD for methane production from different substrates. According to their results, the VFA/TA ratio of < 0.4 signals a stable co-digestion of waste activated sludge and FW or co-digestion of sewage sludge and fruit wastes; however, the VFA/TA ratio of 0.59 may reflect a stable operation of the mono-digestion of FW. In this study, the VFA/TA ratios were lower than 0.5 in all reactors after day 12, suggesting that combination of Air-NBW pre-acclimation and K^+ supplementation could better maintain the pH buffer capacity during the second stage for CH_4 production from high saline FW.

3.3.4. *Changes of TAN and FAN during the second stage*

For the protein-rich substrates, ammonia nitrogen would be released due to the decomposition of proteins, amino acids, urea, and nucleic acids. Considering the addition of KOH in the methanogenic stage to supplement the alkalinity, the conversion of TAN to FAN in a slightly alkaline environment is known as one of the inhibitors to methanogens (Sposob et al., 2020). Thus, the concentrations of TAN and FAN are summarized (Table 3). The TAN and FAN concentrations in the NBW group were slightly higher than those in the corresponding DW group, and the increase in FAN was more obvious in the DW-9.0 and NBW-9.0 reactors because FAN is mainly affected by TAN, pH, and temperature according to Eq. (1). Many previous studies have disclosed the inhibition effects of TAN and FAN on AD, but it is difficult to determine the limit concentration of TAN or FAN to methane production inhibition. Bi et al. (2020) reported that > 3000 mg N/L of TAN would cause reduced methane production and toxicity to methanogens. Dai et al. (2016) and Westerholm et al. (2011), however, found that the methanogenic activity was affected only when the TAN concentration was higher than 5000 mg N/L. In any case, the concentration of TAN in this study (always < 1700 mg N/L) was far lower than the inhibition concentrations as mentioned above, thus no inhibition on methane production was observed.

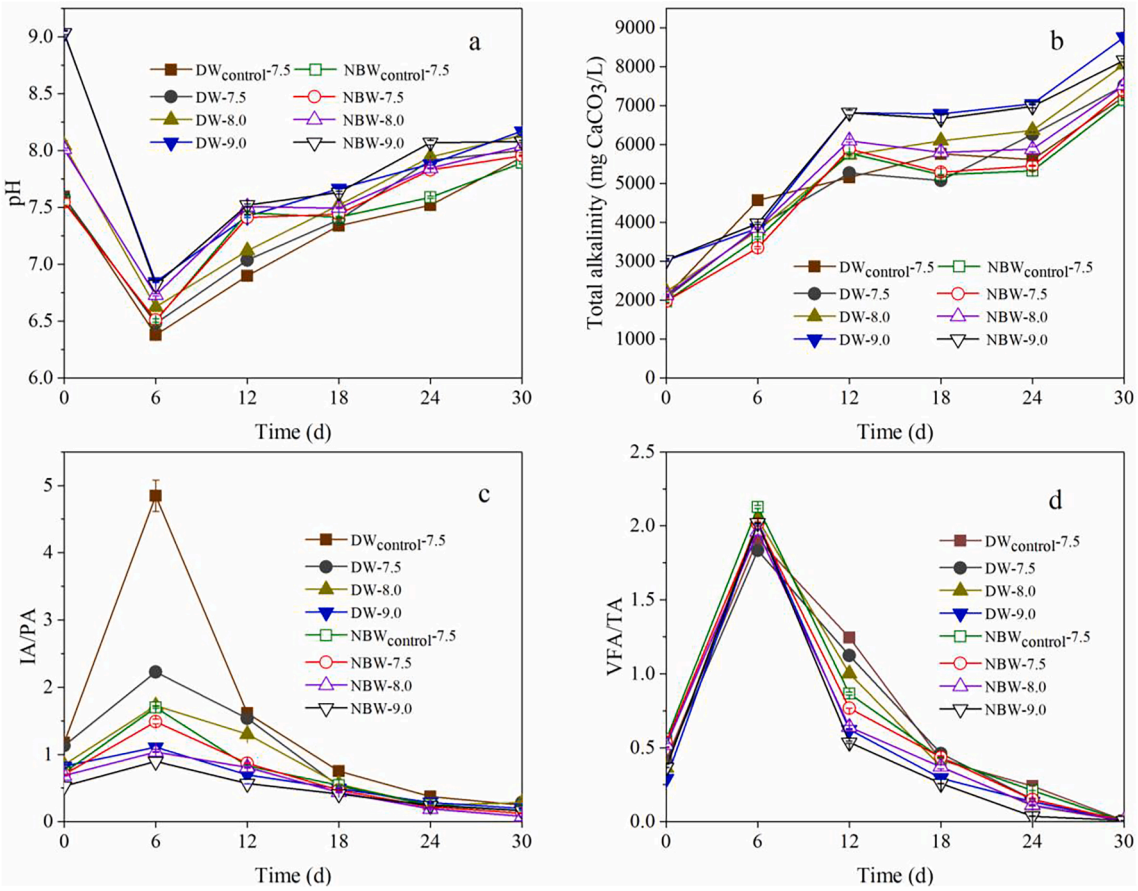


Fig. 6. Profiles of pH (a), total alkalinity (b), IA/PA (c), and VFA/TA (d) during the second stage AD of high saline (20 g NaCl/L) FW.

Table 3
The concentration of TAN and FAN in each reactor during the second stage.

Reactors		DW$_{control}$-7.5	DW-7.5	DW-8.0	DW-9.0	NBW$_{control}$-7.5	NBW-7.5	NBW-8.0	NBW-9.0
TAN (mg N/L)	6 d	1453.80 ± 42.30	1375.9 ± 16.35	1365.15 ± 24.45	1396.80 ± 1.80	1450.05 ± 24.15	1441.05 ± 28.05	1447.65 ± 28.95	1415.85 ± 19.35
	12 d	1367.85 ± 28.35	1393.50 ± 10.50	1438.65 ± 7.95	1566.00 ± 206.10	1347.60 ± 68.10	1410.30 ± 28.2	1610.10 ± 28.20	1552.95 ± 196.35
	18 d	1194.00 ± 38.10	913.05 ± 3.15	1203.75 ± 43.05	1312.2 ± 8.40	1278.00 ± 57.00	1258.20 ± 200.4	1214.70 ± 3.60	1355.55 ± 22.65
	24 d	1213.80 ± 17.70	1075.95 ± 16.35	1065.15 ± 24.45	1182.60 ± 36.00	1329.00 ± 53.10	1396.05 ± 13.05	1410.15 ± 2.55	1415.85 ± 19.35
	30 d	1449.00 ± 11.70	1500.60 ± 29.70	1496.55 ± 14.85	1530.15 ± 25.65	1576.20 ± 17.40	1515.90 ± 20.40	1528.05 ± 13.95	1510.20 ± 15.30
FAN (mg N/L)	6 d	4.81 ± 0.25	5.73 ± 0.20	7.92 ± 0.23	13.39 ± 0.17	6.40 ± 0.33	6.42 ± 0.13	10.67 ± 0.03	12.82 ± 0.18
	12 d	14.88 ± 0.65	20.59 ± 0.08	25.77 ± 0.44	54.83 ± 6.02	50.64 ± 3.68	48.43 ± 0.97	68.21 ± 1.78	68.23 ± 10.10
	18 d	35.06 ± 0.34	30.00 ± 0.77	52.78 ± 3.05	78.90 ± 1.36	44.36 ± 1.49	46.18 ± 7.36	50.36 ± 0.70	75.58 ± 2.90
	24 d	53.18 ± 1.57	109.91 ± 2.81	115.85 ± 6.22	112.41 ± 5.76	67.95 ± 4.20	119.40 ± 1.12	124.48 ± 1.08	197.98 ± 5.14
	30 d	155.82 ± 1.26	182.35 ± 3.77	237.66 ± 4.55	259.87 ± 0.61	152.84 ± 1.49	168.13 ± 3.98	201.26 ± 2.19	215.35 ± 2.07

3.4. Application potential

The research of NBW in the field of AD is becoming a hotspot, however, there are still numerous challenges in practical applications, including increase in the AD reactor volume due to the addition of NBW. This study tried to use Air-NBW to pre-acclimate anaerobically digested sludge that was used as inoculum in the two-stage AD of high salinity FW. On the one hand, results showed that Air-NBW pre-acclimated sludge can significantly enhance the H_2 and CH_4 production, and the reactor volume could be reduced compared with adding Air-NBW directly. On the other hand, the reactor start-up at each stage was accelerated under the high salinity by using Ari-NBW pre-acclimated sludge as inoculum. In addition, from an economic perspective, substituting KOH for NaOH in the methanogenic stage to adjust the initial pH would not increase the cost because it is normally necessary to adjust the pH in the second stage of the two-stage AD of FW. Compared with the addition of KCl to mitigate the inhibition of salt on methanogens, use of KOH can not only achieve higher methane production and

a more stable AD process but also avoid excessive K^+ and Cl^- addition. From this study, the optimal methane production performance was obtained from the NBW-8.0 reactor. Finally, the concentrations of K^+ in the biogas slurry from KOH added reactors (2210.55 ± 53.46, 2484.90 ± 24.40, 3058.05 ± 41.58, 2177.25 ± 49.21, 2625.45 ± 37.55, and 3102.90 ± 30.12 mg/L in the DW-7.5, DW-8.0, DW-9.0, NBW-7.5, NBW-8.0, and NBW-9.0) were higher than those from the reactors without KOH addition (220.65 ± 34.37 and 273.15 ± 69.16 mg/L in the $DW_{control}$-7.5 and $NBW_{control}$-7.5) at the end of the second stage. This indicates that addition of KOH could remarkably improve the fertilizer value of the digestate. Certainly, future research should focus on the application and performance of this strategy on the real high saline FW treatment in the larger-scale and longer-term two-stage AD.

4. Conclusions

This study investigated the performance of two-stage AD of high saline (20 g NaCl/L) FW by using Air-NBW pre-acclimated anaerobically digested sludge and the effects of K^+ addition on methanogenic stage by adding KOH to adjust the initial pH. Results indicate that the H_2 yield increased about 46% in the NBW group, and the addition of KOH also enhanced the CH_4 yield and ETS activity in the second stage. Additionally, a shorter start-up period and more stable performance were obtained in the NBW group with the KOH addition, especially in NBW-8.0, in addition to improved fertilizer value of the digestate.

CRediT authorship contribution statement

Tingting Hou: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Jiamin Zhao:** Data curation. **Zhongfang Lei:** Conceptualization, Formal analysis, Funding acquisition, Supervision, Writing – review & editing. **Kazuya Shimizu:** Formal analysis, Methodology, Supervision. **Zhenya Zhang:** Conceptualization, Project administration, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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